

Chapter 4

Homework 4 due: Thursday 10th April, 2014

All non-starred problems from chapter 4 section 1.

Exercise 3.1. Show that the set of integers, $\mathbb{Z}$, is a group under addition.

Proof. Any integer n can be written as $n = 1 + 1 + \cdots + 1$ for n 1's. So then for any two $m, n \in \mathbb{Z}$ we have

$$\begin{aligned} m + n &= (1 + 1 + \cdots + 1) + (1 + 1 + \cdots + 1) \\ &= 1 + 1 + \cdots + 1 \end{aligned}$$

which is just $m + n$ 1's. So $\mathbb{Z}$ is closed.

$1 + 0 = 0 + 1 = 1$ so then $n + 0 = 1 + 1 + \cdots + 1 + 0 = n$, and 0 is the identity.

$1 + (-1) = (-1) + 1 = 0$. So then the inverse of n is just $-n = -1 - 1 \cdots - 1$.

Associativity follows since addition of any multiple of integers is just adding a list of 1's, and so the order is irrelevant. Thus $(m + n) + m' = m + (n + m')$ for any $n, m, m' \in \mathbb{Z}$. $\square$

Exercise 3.2. Prove that the set of even integers is a group under addition.

Proof. Let $2\mathbb{Z} = \{n \in \mathbb{Z} \text{ such that } 2|n\}$ for 2. Let $a, b \in 2\mathbb{Z}$. Then $a = 2x$ and $b = 2y$ for some $x, y \in \mathbb{Z}$. $a + b = 2x + 2y = 2(x + y)$ so $2|a + b$, and $2\mathbb{Z}$ is closed. The identity is 0, $a^{-1} = -a = -2x$ which is divisible by 2, and associativity follows from associativity of $\mathbb{Z}$. $\square$

Exercise 3.3. Prove that integers of n multiples for $n \in \mathbb{Z}^+$ are a group under addition.

Proof. Let $n\mathbb{Z} = \{x \in \mathbb{Z} \text{ such that } n|x\}$ for some integer n . Let $a, b \in n\mathbb{Z}$. Then $a = nx$ and $b = ny$ for some $x, y \in \mathbb{Z}$. $a + b = nx + ny = n(x + y)$ so $n|a + b$, and $n\mathbb{Z}$ is closed. The identity is 0, $a^{-1} = -a = -nx$ which is divisible by n , and associativity follows from associativity of $\mathbb{Z}$. $\square$

Exercise 3.4. Prove that the set of all 2×2 matrices with determinant 1 and entries from $\mathbb{R}$ is a group under matrix multiplication.

Proof. We'll prove this for the more general case for $n \times n$ matrices. Let $M_{i,j}$ represent the entry in the matrix M at row i and column j . Multiplication of matrices A and B is defined by

$$(AB)_{i,j} = \sum_{k=1}^m A_{i,k} B_{k,j}$$

Where m is the number of rows in B . This product is defined if and only if the number of columns in A equals the number of rows in B . This will always be the case if both A and B are $n \times n$ matrices.

If we previously show that $\mathbb{R}$ is a group under multiplication and addition. So the each entry in AB , $AB_{i,j}$, is a real number. Furthermore from the formula, we can see that if A

and B are $n \times n$ matrices then AB is also a 2×2 matrix. Finally, A and B are square matrices and $\det(A) = \det(B) = 1$. This implies that $\det(AB) = \det(A)\det(B) = 1$, and thus this set is closed under multiplication..

The identity will be the identity matrix,

$$I = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

As

$$\begin{aligned} (AI)_{i,j} &= \sum_{k=1}^m A_{i,k} I_{k,j} = A_{i,1}0 + A_{i,2}0 + \cdots + A_{i,j}1 + \cdots + A_{i,m}0 \\ (IA)_{ij} &= \sum_{k=1}^m I_{k,j} A_{i,k} = 0A_{i,1} + 0A_{i,2} + \cdots + 1A_{i,j} + \cdots + 0A_{i,m} \\ &= A_{i,j} \end{aligned}$$

For each matrix A there exists an A^{-1} such that $AA^{-1} = A^{-1}A = I$ since $\det(A) = 1$.

For associativity

$$\begin{aligned} (AB)_{i,j} C_{ij} &= \left(\sum_{k=1}^m A_{i,k} B_{k,j} \right) C_{ij} = \sum_{k=1}^m \left(\sum_{l=1}^m A_{i,k} B_{k,l} \right) C_{l,j} \\ &= \sum_{k=1}^m \sum_{l=1}^m (A_{i,k} B_{k,l}) C_{l,j} \text{ by distribution} \\ &= \sum_{k=1}^m \sum_{l=1}^m A_{i,k} (B_{k,l} C_{l,j}) \text{ by associativity of } \mathbb{R}. \\ &= A_{i,j} (B_{i,j} C_{i,j}) \end{aligned}$$

□

Exercise 3.5. Show that the following are *not* groups under the given operation.

1. $\mathbb{Z}$ under multiplication.

Proof. $2 \in \mathbb{Z}$, but $2^{-1} \notin \mathbb{Z}$ since $2^{-1}2 = 1 \Rightarrow 2^{-1} = \frac{1}{2}$.

□

2. $S = \{z | z \in \mathbb{Z} \text{ and } z = \sqrt{5}\}$ under any operation.

Proof. $\sqrt{5} \notin \mathbb{Z}$ so then $S = \emptyset$. So S is not a group.

□

3. $\mathbb{Q}$, the set of rationals under multiplication.

Proof. $0 \in \mathbb{Q}$, but 0 has no multiplicative inverse.

□

4. The set of odd integers under addition.

Proof. 3 and 5 are odd, but $3 + 5 = 8 = 2 \cdot 4$ which is not odd. So it is not closed. $\square$

Exercise 3.6. Let $S = \{a + b\sqrt{5} \mid a, b \text{ are not both zero and } a, b \in \mathbb{Q}\}$. Prove that S is a group. (Hint: rationalize the denominator to show that the inverse of $a + b\sqrt{5}$ is in S .)

Proof. Let $x, y \in S$ so then $x = a + b\sqrt{5}$ and $y = c + d\sqrt{5}$ for some $a, b, c, d \in \mathbb{Q}$. $xy = ac + (b + d)\sqrt{5} + 5bd = ab + 5bd + (a + b)\sqrt{5}$ which is in S as $\mathbb{Q}$ is closed under multiplication and addition. 1 is the identity. $x^{-1} = \frac{1}{a + b\sqrt{5}} \frac{(a - b\sqrt{5})}{(a - b\sqrt{5})} = \frac{a - b\sqrt{5}}{a^2 - 5b^2} = \frac{a}{a^2 - 5b^2} + \frac{-b}{a^2 - 5b^2}\sqrt{5} \in S$. Associativity follows by the associativity of $\mathbb{R}$. $\square$

Exercise 3.7. Show that $\{1, 2, 3\}$ under multiplication modulo 4 is *not* a group but that $\{1, 2, 3, 4\}$ under multiplication modulo 5 is a group. (Example: $2 \cdot 3 = 6$ and $6 = 2 \pmod{4}$ so $[6] = [2]$ and thus $2 \cdot 3$ is in $\{1, 2, 3\}$).

Proof. $2 \cdot 2 = 0 \pmod{4}$ and $0 \notin \{1, 2, 3\}$ so it is not a group.

		1	2	3	4
	1	1	2	3	4
$\{1, 2, 3, 4\}$:	2	2	4	1	3
	3	3	1	4	2
	4	4	1	2	3

So it is closed, the identity is 1, each element has an inverse, and associativity follows by the associativity of modular multiplication. $\square$

Exercise 3.8. Addition in $\mathbb{Z}_n$ is defined as $[a] + [b] = [a + b]$. Multiplication in $\mathbb{Z}_n$ is defined as $[a] \cdot [b] = [a \cdot b]$. For example, $\mathbb{Z}_4 = \{[0], [1], [2], [3]\}$ for convenience we write $\mathbb{Z}_4 = \{0, 1, 2, 3\}$. $2 + 3 = 5$ and $5 = 1 \pmod{4}$ so $[5] = [1]$. Show that $\mathbb{Z}_2 \times \mathbb{Z}_3$ is a group under modular addition.

Proof. Closure: If $(a, x), (b, y) \in \mathbb{Z}_2 \times \mathbb{Z}_3$ then $(a, x) + (b, y) = (a + b, x + y)$. $\mathbb{Z}_2$ and $\mathbb{Z}_3$ are both groups so $a + b \in \mathbb{Z}_2$ and $x + y \in \mathbb{Z}_3$ thus $(a + b, x + y) \in \mathbb{Z}_2 \times \mathbb{Z}_3$.

Identity: The identity in both $\mathbb{Z}_2$ and $\mathbb{Z}_3$ is 0. Thus $(a, b) + (0, 0) = (a + 0, b + 0) = (a, b)$. Similarly $(0, 0) + (a, b) = (a, b)$. So $(0, 0)$ is the identity in $\mathbb{Z}_2 \times \mathbb{Z}_3$.

Inverse: The inverses of a and b is $\square$

Exercise 3.9. Prove that the set of all $n \times n$ matrices with real entries is an abelian group under addition.

Proof. Let $M_{i,j}$ represent the entry in the matrix M at row i and column j . Let A and B be $n \times n$ matrices then the entry in the i^{th} row and j^{th} column is found as follows:

$$(A + B)_{i,j} = A_{i,j} + B_{i,j}$$

$A + B$ is then well defined as long as they have matching dimension. Since the real numbers are a group under addition with identity 0 and $a^{-1} = -a$ for any $a \in \mathbb{R}$, the addition of

entries is closed, has identity 0, inverse $-A_{ij}$, and is associative. Also the addition of entries is abelian since $\mathbb{R}$ is abelian.

Since closure, associativity, and commutativity hold for each entry it holds for matrices with real entries. The identity is just a matrix with all zeros,

$$\mathbf{0} = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}$$

and the inverse for a matrix A is just $-A$, e.g.,

$$\text{if } A = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & & a_{2,n} \\ \vdots & & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,n} \end{bmatrix} \text{ then } A^{-1} = \begin{bmatrix} -a_{1,1} & -a_{1,2} & \cdots & -a_{1,n} \\ -a_{2,1} & -a_{2,2} & & -a_{2,n} \\ \vdots & & \ddots & \vdots \\ -a_{n,1} & -a_{n,2} & \cdots & -a_{n,n} \end{bmatrix}$$

□

Exercise 3.10. Prove that the set of all 2×2 matrices with determinant 1 and entries from $\mathbb{R}$ is a non-abelian group under matrix multiplication (we already know that it is a group by exercise 3.4).

Proof. Let

$$A = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$$

then

$$AB = \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix} \text{ and } BA = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

So $AB \neq BA$, and this group is not abelian. □

Exercise 3.11. ★

Exercise 3.12. Prove that G is an abelian group if and only is $(a \oplus b)^{-1} = a^{-1} \oplus b^{-1}$ for all $a, b \in G$.

Proof. ($\Rightarrow$) Suppose that G is abelian. $(a \oplus b)^{-1} = b^{-1} \oplus a^{-1}$ since $b^{-1} \oplus a^{-1} \oplus (a \oplus b) = 0_G$ and inverses are unique. Since G is assumed to be abelian

$$(a \oplus b)^{-1} = b^{-1} \oplus a^{-1} = a^{-1} \oplus b^{-1}.$$

($\Leftarrow$) Suppose that $(a \oplus b)^{-1} = a^{-1} \oplus b^{-1}$. Similarly to above $(a \oplus b)^{-1} = b^{-1} \oplus a^{-1}$ so then

$$\begin{aligned} (a \oplus b) \oplus (a \oplus b)^{-1} &= 0_G \\ a \oplus b \oplus (a^{-1} \oplus b^{-1}) &= 0_G \\ a \oplus b \oplus (a^{-1} \oplus b^{-1}) \oplus (b \oplus a) &= 0 \oplus (b \oplus a) \\ a \oplus b \oplus (a^{-1} \oplus \cancel{b^{-1} \oplus b}) \oplus a &= b \oplus a \\ a \oplus b \oplus \cancel{(a^{-1} \oplus a)} &= b \oplus a \\ a \oplus b &= b \oplus a \end{aligned}$$

This is easier to see using juxtaposition to denote the operation instead of “ $\oplus$ ”.

($\Rightarrow$) Suppose that G is abelian. Then $(ab)^{-1} = (ba)^{-1}$. $(ba)^{-1}ba = e \Rightarrow (ba)^{-1} = a^{-1}b^{-1}$ as required.

($\Leftarrow$) Suppose that $(ab)^{-1} = a^{-1}b^{-1}$. Similarly to above $(ab)^{-1} = b^{-1}a^{-1}$. So then $ab(ab)^{-1} = e \Rightarrow ab(a^{-1}b^{-1}) = e \Rightarrow abab^{-1}ba = ba \Rightarrow ab = ba$ $\square$

Exercise 3.13. Suppose that if G is a group such that if $a, b, c \in G$ and $a \oplus b = c \oplus a$ then $b = c$. Prove that G is an Abelian group, i.e., that $a \oplus b = b \oplus a$ for all $a, b \in G$. Hint: consider the element $a \oplus (b \oplus a) = (a \oplus b) \oplus a$.

Proof. Let $a, b, c \in G$ and suppose that $a \oplus b = c \oplus a$ implies $b = c$. G is a group so then $a \oplus (b \oplus a) = (a \oplus b) \oplus a$ by associativity. By the assumption $a \oplus (b \oplus a) = (a \oplus b) \oplus a$ implies that $a \oplus b = b \oplus a$ so then G is abelian. $\square$